

Navigators — Technical Terms Cheat Sheet

Quick revision • simple meaning + role in the project

Navigation & Sensors

Term	Full Form	What it does
GNSS	Global Navigation Satellite System	Satellite-based absolute positioning.
GPS	Global Positioning System	A US GNSS constellation.
NavIC	Navigation with Indian Constellation	India's regional satellite navigation system.
IMU	Inertial Measurement Unit	Motion sensors, mainly accelerometer + gyroscope.
Accelerometer	—	Measures linear acceleration.
Gyroscope	—	Measures angular velocity and helps estimate rotation.
Magnetometer	—	Measures magnetic field; useful for heading information.
6-DOF	Six Degrees of Freedom	3-axis acceleration + 3-axis angular motion.
9-DOF	Nine Degrees of Freedom	Accelerometer + gyroscope + magnetometer.

Navigation Engine

Term	Full Form	What it does
INS	Inertial Navigation System	Uses IMU data to estimate orientation, velocity and position.
DR	Dead Reckoning	Continues position estimation from the last known state and motion.
EKF	Extended Kalman Filter	Fuses noisy sensor/AI measurements into a state estimate.
NHC	Non-Holonomic Constraint	Uses vehicle motion constraints to reduce navigation drift.
Attitude	—	Orientation: roll, pitch and yaw.
Sensor Fusion	—	Combines multiple sensor/AI sources.
Map Matching	—	Matches estimated position to the most likely road.
OSM	OpenStreetMap	Map/road data useful for offline map matching.

AI / ML

Term	Full Form	What it does
AI	Artificial Intelligence	Broad field of machine-based intelligent systems.
ML	Machine Learning	Learns patterns from data for prediction.
AI Speed Estimation	—	Estimates vehicle speed from smartphone motion sensors.
Training	—	Model learns from sensor data and target values.
Inference	—	Trained model predicts from new sensor data.
Ground Truth	—	Reference value used as target/evaluation truth.
Sliding Window	—	Uses short consecutive sensor-data segments.
TCN	Temporal Convolutional Network	Neural network for temporal patterns.
LSTM	Long Short-Term Memory	Recurrent network for sequential data.
Data Leakage	—	Target/test information improperly reaches model inputs/training.

Vehicle & Data

Term	Full Form	What it does
OBD-II	On-Board Diagnostics II	Vehicle diagnostic interface; may provide vehicle data.
CAN	Controller Area Network	Communication network between vehicle electronic units.
Bias	—	Systematic sensor offset that can cause drift.

Term	Full Form	What it does
Noise	—	Random variation in measurements.
Filtering	—	Reduces unwanted sensor noise.
Coordinate Frame	—	Reference axes of phone/vehicle/navigation system.
Velocity	—	Speed with direction.
Speed	—	Magnitude of motion, e.g. 60 km/h.

Deployment & Evaluation

Term	Full Form	What it does
ONNX	Open Neural Network Exchange	Portable/interoperable ML model format.
ONNX Runtime	—	Runs ONNX models locally, including Android.
Edge AI	—	AI inference performed locally on the device.
MAE	Mean Absolute Error	Average absolute prediction error; lower is better.
RMSE	Root Mean Square Error	Penalizes larger errors more; lower is better.
Train Set	—	Data used to train the model.
Validation Set	—	Data used for development/tuning.
Test Set	—	Held-out data for final evaluation.
IO-VNBD	Inertial and Odometry Vehicle Navigation Benchmark Dataset	Real driving benchmark data for training/evaluation.

Whole system: GNSS gives absolute position → IMU gives motion → AI estimates vehicle speed → INS/EKF fuses information → NHC/map matching reduce drift → ONNX Runtime runs AI locally.